

Starter Project Outline: Quantum Active-Space Astrochemistry for $\text{H} + \text{H}_2\text{CO}$ on Ice

QDYNAMICS Project Notes

July 2, 2026

1 Core Idea

The first chemically meaningful project should be simple enough to implement with the current QDYNAMICS/Psi4 pipeline, but relevant enough to grow into the larger astrochemistry program. The proposed starting reaction is hydrogen addition to formaldehyde,



This reaction is a natural first target because it is small, radical-like, relevant to grain-surface methanol chemistry, and has competing product channels. The project can start in the gas phase, then add water ice, then reduce the problem to an active-space Hamiltonian suitable for exact diagonalization and ADAPT-VQE.

2 Why This System

- It is astrochemically motivated: hydrogenation of $\text{CO}/\text{H}_2\text{CO}$ is central to methanol and complex organic molecule formation on dust-grain ice.
- It is small enough for a first active-space workflow.
- It has two channels, so the final observable can be a branching ratio rather than only a ground-state energy.
- It naturally introduces tunneling, because H motion controls the low-temperature rate.
- It can be embedded gradually in water: first H_2O , then $(\text{H}_2\text{O})_4$, then $(\text{H}_2\text{O})_8$.

3 Computational Ladder

The project should proceed through a ladder where each stage adds only one new complication.

Stage	System	Purpose
0	H ₂	Verify Psi4, AO/MO integrals, spin-orbital indexing.
1	H ₂ CO	Verify molecular integrals for the first real target molecule.
2	H + H ₂ CO	Build reactant/product and stretched-bond geometries.
3	H + H ₂ CO + H ₂ O	Add the first ice hydrogen-bond field.
4	H + H ₂ CO + (H ₂ O) _n	Move toward ice clusters, $n = 4, 8, 16$.
5	Active-space Hamiltonian	Downfold to CAS(N_e, N_o) for quantum solvers.

4 Hamiltonian Target

For a fixed nuclear geometry R , the electronic Hamiltonian in spin orbitals is

$$\hat{H}_{\text{el}}(R) = E_{\text{nuc}}(R) + \sum_{pq} h_{pq}(R) a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs}(R) a_p^\dagger a_q^\dagger a_s a_r. \quad (3)$$

The current Psi4 pipeline is already moving in this direction by constructing

$$h_{pq} \quad \text{and} \quad h_{pqrs} \quad (4)$$

from AO integrals and MO coefficients.

In chemist notation, the spatial two-electron integral is

$$(ij|kl) = \int d\mathbf{r}_1 d\mathbf{r}_2 \phi_i(\mathbf{r}_1) \phi_j(\mathbf{r}_1) \frac{1}{r_{12}} \phi_k(\mathbf{r}_2) \phi_l(\mathbf{r}_2). \quad (5)$$

In spin-orbital form,

$$\langle pq|rs \rangle = (ij|kl) \delta_{\sigma_p \sigma_q} \delta_{\sigma_r \sigma_s}. \quad (6)$$

This is why the spin blocks in the code require the first pair to share spin and the second pair to share spin.

5 First Reaction Geometry Set

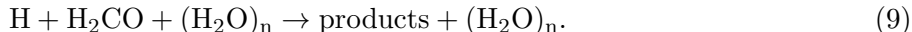
Before locating a full transition state, use a small controlled geometry set:

1. Reactant-like geometry: H far from the carbonyl carbon or oxygen.
2. Midpoint geometry: H approaching either C or O along a chemically reasonable coordinate.
3. Product-like geometry: CH₃O or CH₂OH radical geometry.

This gives an initial reaction profile without yet needing a full nudged elastic band or transition-state search.

6 Ice Extension

The ice environment should be introduced gradually:



The first target observable is the ice-induced barrier shift,

$$\Delta\Delta E_{\text{ice}}^{\ddagger} = \Delta E_{\text{ice}}^{\ddagger} - \Delta E_{\text{gas}}^{\ddagger}. \quad (10)$$

If this quantity is negative, the ice lowers the barrier. If it is positive, the ice suppresses the channel.

7 Active-Space Reduction

The full molecular orbital space will become too large once water molecules are added. The project therefore needs active-space reduction. The first chemically motivated active orbitals are:

- carbonyl π orbital,
- carbonyl π^* orbital,
- incoming H 1s orbital,
- radical/product SOMO,
- oxygen lone pair if hydrogen bonding or proton-coupled effects become important.

A plausible first active space is

$$\text{CAS}(5e, 5o) \quad \text{or} \quad \text{CAS}(7e, 7o), \quad (11)$$

corresponding to 10 or 14 spin orbitals before qubit tapering.

7.1 Active-Space Selection Strategy

The first implementation can be manual and chemistry-driven. Later, it should become automatic:

1. Run a mean-field calculation for each geometry.
2. Inspect orbital characters near the HOMO/LUMO window.
3. Include orbitals involved in forming or breaking bonds.
4. Add natural orbital occupation analysis when correlated references are available.
5. Freeze inactive occupied orbitals and discard high-energy virtual orbitals.

8 Validation Targets

For every geometry, validate the electronic data before running a quantum algorithm:

- Check electron count from the density matrix.
- Check one-electron energy.
- Check two-electron Hartree-Fock energy:

$$E_{2e}^{\text{HF}} = \frac{1}{2} \sum_{pq} n_p n_q [(pp|qq) - (pq|qp)]. \quad (12)$$

- Compare active-space exact diagonalization with the ADAPT-VQE result.

9 Near-Term Software Tasks

1. Keep H_2 as the regression test for integral construction.
2. Add a TESTS case for isolated H_2CO .
3. Add a TESTS case for $\text{H} + \text{H}_2\text{CO}$ at one reactant-like geometry.
4. Add a data object that stores E_{nuc} , h_{pq} , h_{pqrs} , MO coefficients, occupations, and metadata.
5. Add active-space slicing of h_{pq} and h_{pqrs} .
6. Add Jordan-Wigner mapping for the active-space molecular Hamiltonian.
7. Validate against exact diagonalization before connecting to ADAPT-VQE.

10 First Publishable Arc

The first paper-level story can be:

Adaptive quantum active-space simulations of hydrogen addition to formaldehyde, with gas-phase benchmarks and first water-ice barrier shifts.

The first results should include:

- gas-phase reaction profile for CH_3O and CH_2OH channels,
- active-space Hamiltonian construction and validation,
- exact active-space and ADAPT-VQE energies,
- first comparison between gas phase and one-water or small-cluster ice embedding,
- resource estimates as a function of active-space size.

11 Immediate Next Step

The next concrete system after H_2 is isolated formaldehyde, H_2CO . This should be used to test the Psi4 integral pipeline for a real astrochemical molecule before adding the incoming H atom or ice.